

## Multiple Choice (10 marks)

1. Statement 1| Word 2 Vec parameters were not initialized using a Restricted Boltzman Machine. Statement 2| The tanh function is a nonlinear activation function.
  - (a) True, True
  - (b) False, False
  - (c) True, False
  - (d) False, True
2. Statement 1| The BLEU metric uses precision, while the ROGUE metric uses recall. Statement 2| Hidden markov models were frequently used to model English sentences.
  - (a) True, True
  - (b) False, False
  - (c) True, False
  - (d) False, True
3. Averaging the output of multiple decision trees helps \_\_.
  - (a) Increase bias
  - (b) Decrease bias
  - (c) Increase variance
  - (d) Decrease variance
4. Statement 1| Overfitting is more likely when the set of training data is small. Statement 2| Overfitting is more likely when the hypothesis space is small.
  - (a) True, True
  - (b) False, False
  - (c) True, False
  - (d) False, True
5. The numerical output of a sigmoid node in a neural network:
  - (a) Is unbounded, encompassing all real numbers.
  - (b) Is unbounded, encompassing all integers.
  - (c) Is bounded between 0 and 1.
  - (d) Is bounded between -1 and 1.

## True / False (10 marks)

*Answer True or False*

6. True or False: Knowing  $P(E, F)$ ,  $P(H)$ , and  $P(E|H)$  is sufficient to calculate  $P(H|E, F)$ .
7. True or False: The original ResNets and Transformers are indeed feedforward neural networks.
8. True or False: Fully connected networks are the preferred choice for high-resolution image classification due to their superior performance.
9. True or False: Linear hard-margin SVM can only be effectively applied when the training data is linearly separable.
10. True or False: The variance of the Maximum A Posteriori (MAP) estimate is the same as that of the Maximum Likelihood Estimate (MLE).

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the computational cost associated with performing a single gradient descent update given a stored gradient vector  $g$ . How does this relate to the dimensionality  $D$  of the problem?
12. Discuss how unsupervised learning can be applied to analyze a large dataset of medical records from heart disease patients to identify distinct patient clusters for tailored treatments.

*End of Paper*